

Unit 4: Major Concepts Regarding Research Design (8 Hours)

4.2 Qualitative vs. Quantitative Research Methods

- **Qualitative Research:**

Focuses on understanding human behavior, experiences, and meanings. It uses non-numerical data like words, interviews, and observations.

- Examples: Case studies, interviews, focus groups.
- Data type: Descriptive and interpretive.
- Goal: To explore and understand phenomena.

- **Quantitative Research:**

Focuses on numerical measurement and statistical analysis to test hypotheses and identify patterns.

- Examples: Surveys, experiments, statistical modeling.
- Data type: Numerical and measurable.
- Goal: To test relationships and make predictions.

4.3 Population and Sampling Procedure

- **Population:**

The complete set of individuals, items, or data relevant to a research study (e.g., all university students in Nepal).

- **Sample:**

A smaller subset of the population chosen to represent the whole group.

- **Sampling Techniques:**

- *Probability Sampling:* Every member has an equal chance (e.g., simple random, stratified, cluster sampling).
- *Non-Probability Sampling:* Selection based on researcher's judgment (e.g., purposive, convenience sampling).

- **Sample Size:**

Should be large enough to represent the population and achieve statistical validity.

4.4 Tools and Techniques for Data Collection

Common tools for gathering data include:

- **Surveys/Questionnaires:** For collecting structured responses.
- **Interviews:** Personal interaction to gather detailed insights.

- **Experiments:** Used to test causal relationships.
- **Checklists/Observation Sheets:** For recording behavioral or environmental data.

4.5 Use of Statistical Measures

Statistical tools help summarize and interpret data:

- **Measures of Central Tendency:** Mean, Median, Mode.
- **Measures of Dispersion:** Range, Variance, Standard Deviation.
- **Measures of Relationship:** Correlation and Regression for analyzing associations between variables.

4.6 Interpreting Data and Drawing Conclusions

Data interpretation involves:

- **Organizing data** into frequency tables, charts, or graphs.
- **Analyzing trends and relationships** statistically.
- **Drawing conclusions** based on results and relating them to hypotheses and objectives.

4.7 Ethical Considerations and Approvals

Ethical research ensures:

- **Informed consent** from participants.
- **Confidentiality** and privacy of data.
- **Avoidance of plagiarism and bias.**
- **Approval from institutional review boards (IRB)** or ethics committees before data collection.

4.8 Findings and Discussion

- **Findings:** Presentation of analyzed data, often with tables and figures.
- **Discussion:** Interpretation of findings in light of research questions, theories, and previous studies.
- It explains **what the results mean** and **how they contribute** to existing knowledge.